Homework III

Title：Emergency Room Patient Management System (Max-Heap)

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Problem Description

In a hospital Emergency Room (ER), patients must be treated based on the severity of their condition (ranked from 1 to 100). You are required to implement a Max-Heap to manage the patient queue efficiently.

Requirements

Data Structure: Implement a Max-Heap to store patient records.
Time Complexity: Insert and Extract-Max must be completed in O(lg n) time.

Submission:

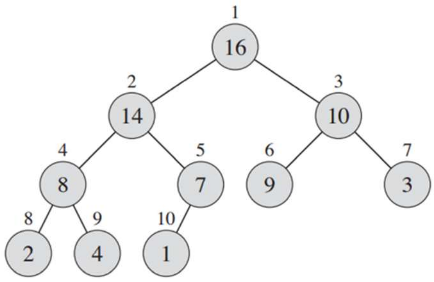
- Source code.
- Execution screenshots.
- A brief report explaining the heapify process (insertion, extraction, and update).
- The Output file (according to the input file: input.txt)

Command Definitions

The system should read commands line by line:

Command	Action	Description
I <name> <priority>	Insert	Add a new patient. name is a string (no spaces); priority is an integer (1–100).
E	Extract-Max	Remove and return the patient with the highest priority.
U <index> <new_priority>	Update	Update the priority of the patient at the specified 1-based index.
S	Show	Display the current heap in array index order (1, 2, 3...).
Q	Quit	Terminate the program.

Example to show max-heap: (following index order: 1, 2, 3, 4, 5, 6, …, 9, 10) [(16), (14), (10), (8), (7), (9), (3), (2), (4), (1)]



Example:

INPUT

I Alice 10
I Bob 25
I Cindy 15
S
E
I David 30
U 1 40
S
E
Q

OUTPUT

[S] Current Heap: [(Bob, 25), (Alice, 10), (Cindy, 15)]
[E] Extract: Bob (25)
[S] Current Heap: [(David, 40), (Cindy, 15), (Alice, 10)]
[E] Extract: David (40)